

CHE 201 Fall 2019 HW 5

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TOTAL POINTS

64 / 100

QUESTION 1

11 15 / 15

- ✓ - **0 pts** Correct
- **8 pts** The v_1 & v_2 are incorrect
- **7 pts** The initial and final mass were wrong

QUESTION 2

22 15 / 15

- ✓ - **0 pts** Correct
- **4 pts** Missing comments if work is done by or on the water vapor
- **4 pts** Missing identification of work as an area on the diagram
- **2 pts** Sketch 1 is wrong. $P_2 > P_1$
- **1.5 pts** Sketch 2 is wrong. $P_3 > P_1$
- **1.5 pts** Sketch 3 is wrong. $V_4 > V_1$
- **2 pts** Sketch 4 is wrong. $P_5 < P_1$
- **1 pts** Work 1-2 is the whole area under the curve
- **1 pts** Work 1-5 is the whole area under the curve
- **1 pts** Process 1-2 should be a curved line
- **1 pts** Process 1-5 should be a curved line
- **1 pts** Missing the process line on the diagram for process 1-3
- **1 pts** Missing the process line on the diagram for process 1-4
- **4 pts** Missing sketches for processes 1-3 and 1-4
- **1 pts** The arrow in process 1-5 should be pointing down to indicate the direction of the process
- **2 pts** The initial state is water vapor and not saturated liquid water
- **2 pts** Missing comments about work for processes 1-3 and 1-4

QUESTION 3

33 0 / 20

- **0 pts** Correct
- **4 pts** Did not multiply specific volume by mass
- **3 pts** The equation used for work would result in a negative value. It should be $(P_1V_1 - P_2V_2)/(n-1)$ or $(P_2V_2 - P_1V_1)/(1-n)$
- **20 pts** Missing answer
- **7 pts** Should have used the polytropic process relationship $PV^n = \text{constant}$ to calculate the volume in state 2
- **7 pts** Wrong calculation of the integral for work. The correct answer should be $W = 31.4 \text{ Btu}$ (work done by the system).
- ✓ - **20 pts** Wrong Solution. The correct answer should be $W = 31.4 \text{ Btu}$ (work done by the system).
- **3 pts** Wrong calculation/unit conversion. The correct answer should be $W = 31.4 \text{ Btu}$ (work done by the system).

QUESTION 4

44 15 / 15

- ✓ - **0 pts** Correct
- **1 pts** $v_a \approx v_f(400^\circ\text{F}) = 0.01864 \text{ ft}^3/\text{lb}$ OR $v_a = 0.018334 \text{ ft}^3/\text{lb}$ (from Table A-5E)
- **1 pts** $h_a = h_f(400^\circ\text{F}) + v_f(400^\circ\text{F})[P - P_{\text{sat}}(400^\circ\text{F})] = 384.6 \text{ Btu/lb}$ OR $h_a = 378.50 \text{ Btu/lb}$ (from Table A-5E)
- **1 pts** Using Table A-10E and Eq 3.11, $v_b \approx v_f(95^\circ\text{F}) = 0.01371 \text{ ft}^3/\text{lb}$
- **1 pts** Using Table A-10E and Eq 3.14, $h_b \approx h_f(95^\circ\text{F}) = 42.47 \text{ Btu/lb}$
- **1 pts** Using Table A-13 and Eq 3.11, $v_c \approx v_f(20^\circ\text{C}) = 1.6386 \times 10^{-3} \text{ m}^3/\text{kg}$
- **1 pts** Using Table A-13 and Eq 3.14, $h_c \approx h_f(20^\circ\text{C}) = 274.26 \text{ kJ/kg}$
- **1 pts** Using Table A-17 and Eq 3.11, $v_d \approx v_f(0^\circ\text{C}) = 1.89 \times 10^{-3} \text{ m}^3/\text{kg}$

- **1 pts** Using Table A-17 and Eq 3.14, $h_d \approx h_f (0^\circ\text{C}) =$

95.1 kJ/kg

- **15 pts** Missing answer

- **1 pts** Missing v.a)

- **4 pts** Missing answer for d)

- **2 pts** Missing location of points b and c

- **15 pts** Missing answer

- **3 pts** Wrong answer for a)

- **4 pts** Missing answer for c)

- **7 pts** Wrong or missing T-v diagram sketches with the location of the points

- **10 pts** Energy balance equation was incorrect

QUESTION 5

5 5 4 / 20

- **0 pts** Correct

- **2 pts** Wrong or missing p-v diagram with sketch of the cycle

- **5 pts** Missing or wrong calculation of Q_{in}

✓ - **3 pts** Missing or wrong calculation of thermal efficiency of the cycle. Correct answer is 12%

✓ - **5 pts** Missing or wrong calculation of W_{cycle} or Q_{out}

✓ - **2.5 pts** Missing calculation of V_3 and V_1 OR V_3 and T_3

- **1 pts** W_{3-1} is negative

✓ - **1.5 pts** Correct equation but wrong solution for thermal efficiency. Correct answer is 12%

- **20 pts** Missing answer

- **1 pts** $T_3 = 887\text{K}$

✓ - **2.5 pts** Missing or wrong calculation of T_2 , T_3 and P_2

✓ - **1.5 pts** Did not show conversion to english units

- **10 pts** Correct solution process, but wrong values.

The correct answer for part a is 12%.

QUESTION 6

6 6 15 / 15

✓ - **0 pts** Correct

- **15 pts** Prob 3.61

- **15 pts** Final T ?

- **5 pts** T_f Calculation was incorrect

- **5 pts** Specific heat of copper incorrect

1) Homework 5

3.16

$$v_1 = 0.015089 \text{ m}^3/\text{kg}$$

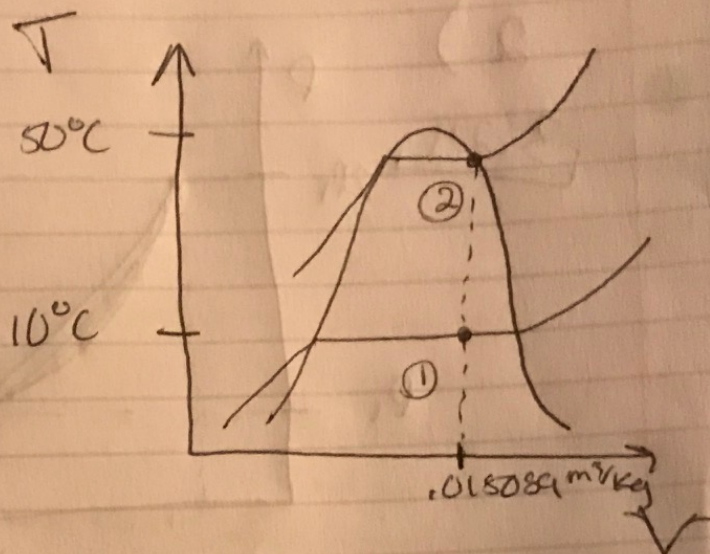
$$v_2 = 0.015089 \text{ m}^3/\text{kg}$$

$$v_1 = v_2$$

$$v_f = 0.015089 =$$

$$v_g = 0.049403 \text{ m}^3/\text{kg}$$

~~$$0.015089 = 0.000793 + x(0.049403 - 0.000793)$$~~



$$0.01505 = 0.000793 + x(0.049403 - 0.000793)$$

$$x = 0.294$$

$$\frac{V}{m} = \frac{1.5 \text{ m}^3}{0.015089} = 99.41 \text{ kg} = \text{total mass}$$

~~$$99.41 \text{ kg}$$~~

$$x = \frac{\text{mass vapor}}{\text{total mass}}$$

$$0.294 \cdot 99.41 \text{ kg} = 29.23 \text{ kg}$$

State 1	Mass of vapor = 29.23 kg.
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$\Rightarrow$ Mass of vapor state 2 = 99.41 kg.

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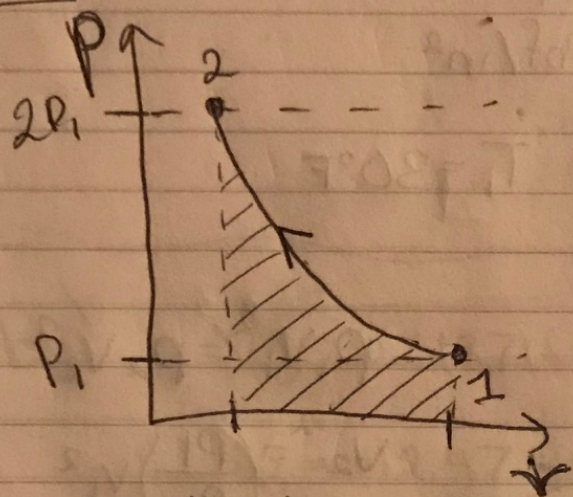
✓ - 0 pts Correct

- 8 pts The v_1 & v_2 are incorrect

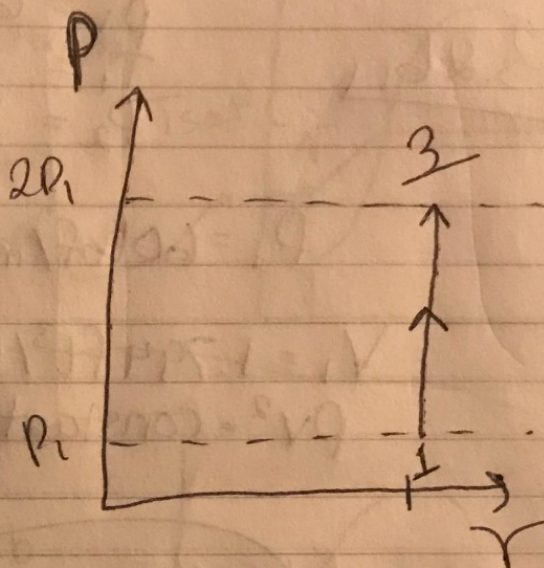
- 7 pts The initial and final mass were wrong

2.)

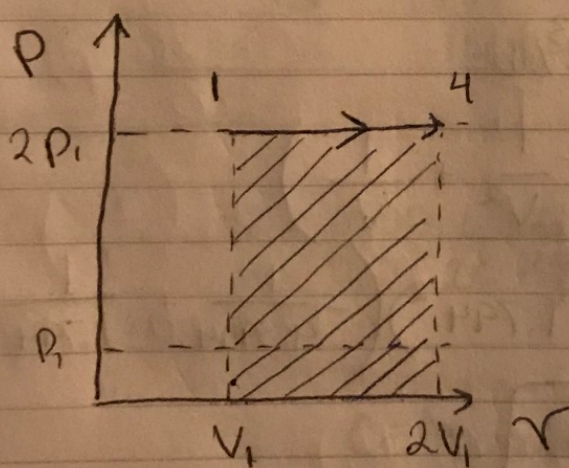
3.24



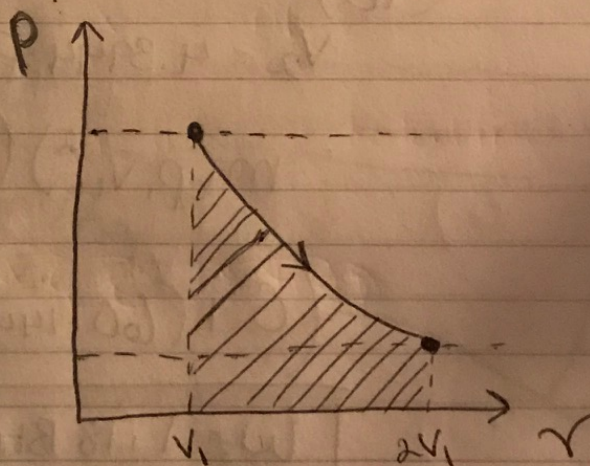
work done on
vapor



work = 0



work done by
vapor



work done by
vapor

→

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✓ - 0 pts Correct

- 4 pts Missing comments if work is done by or on the water vapor
- 4 pts Missing identification of work as an area on the diagram
- 2 pts Sketch 1 is wrong. $P_2 > P_1$
- 1.5 pts Sketch 2 is wrong. $P_3 > P_1$
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- 1 pts Work 1-2 is the whole area under the curve
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- 1 pts Process 1-2 should be a curved line
- 1 pts Process 1-5 should be a curved line
- 1 pts Missing the process line on the diagram for process 1-3
- 1 pts Missing the process line on the diagram for process 1-4
- 4 pts Missing sketches for processes 1-3 and 1-4
- 1 pts The arrow in process 1-5 should be pointing down to indicate the direction of the process
- 2 pts The initial state is water vapor and not saturated liquid water
- 2 pts Missing comments about work for processes 1-3 and 1-4

3.)

3.26

$$m = 0.1 \text{ lb}$$

$$p_1 = 60 \text{ lbf/in}^2$$

$$T_1 = 30^\circ \text{F}$$

$$p_2 = 10 \text{ lbf/in}^2$$

$$p_1 = 60 \text{ lbf/in}^2 \quad T_1 = 30^\circ \text{F}$$

$$V_1 = 1.794 \text{ ft}^3/\text{lb}$$

$$pV^2 = \text{constant}$$

$$p_1 V_1^2 = p_2 V_2^2$$

$$V_2^2 = \left(\frac{p_1}{p_2} \right) V_1^2$$

$$\left(\frac{60}{10} \right) 1.794^2 = 19.3106$$

$$V_2 = 4.3944 \text{ ft}^3/\text{lb}$$

$$m(p_1 V_1^2) \left(-\frac{1}{V_2} + \frac{1}{V_1} \right)$$

$$\left(0.1 (60 \cdot 144 \cdot 1.794^2) \left(-\frac{1}{4.3944} + \frac{1}{1.794} \right) \right) \frac{1}{778}$$

$$W = 1.18 \text{ Btu}$$

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- 0 pts Correct

- 4 pts Did not multiply specific volume by mass

- 3 pts The equation used for work would result in a negative value. It should be $(P_1V_1 - P_2V_2)/(n-1)$ or $(P_2V_2 - P_1V_1)/(1-n)$

- 20 pts Missing answer

- 7 pts Should have used the polytropic process relationship $PV^n = \text{constant}$ to calculate the volume in state 2

- 7 pts Wrong calculation of the integral for work. The correct answer should be $W = 31.4 \text{ Btu}$ (work done by the system).

✓ - 20 pts **Wrong Solution. The correct answer should be $W = 31.4 \text{ Btu}$ (work done by the system).**

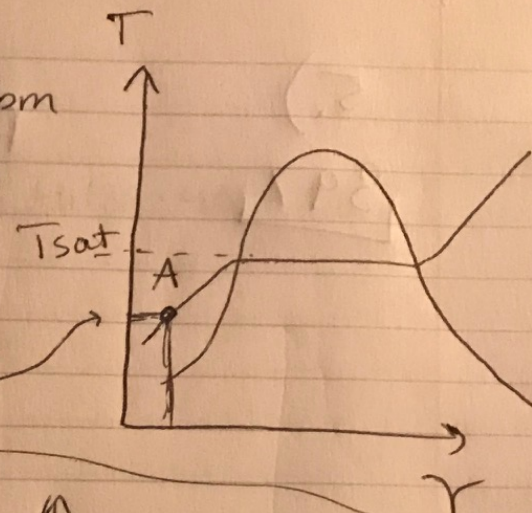
- 3 pts Wrong calculation/unit conversion. The correct answer should be $W = 31.4 \text{ Btu}$ (work done by the system).

4.)

3.31

a.) $V_A = 0.018334 \text{ ft}^3/\text{lbm}$
 $h_A = 378.80 \text{ Btu/lb}$

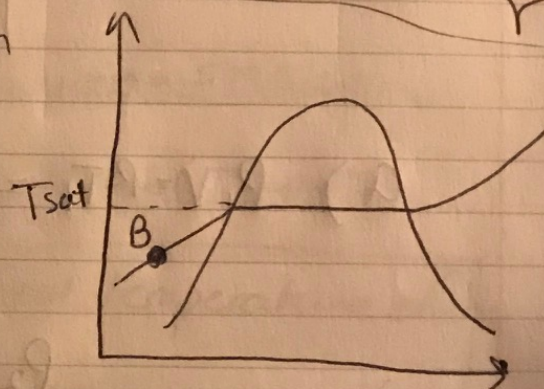
A = 400°F
 3000 lbf/in^2



b.) $V_B = 0.01371 \text{ ft}^3/\text{lbm}$

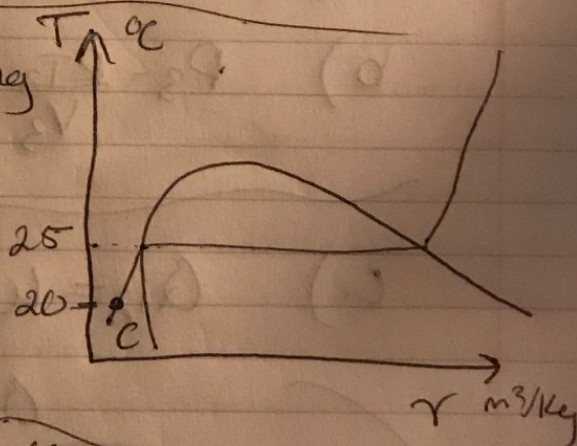
$h_B = 42.47 \text{ Btu/lb}$

B = 95°F
 150 lbf/in^2



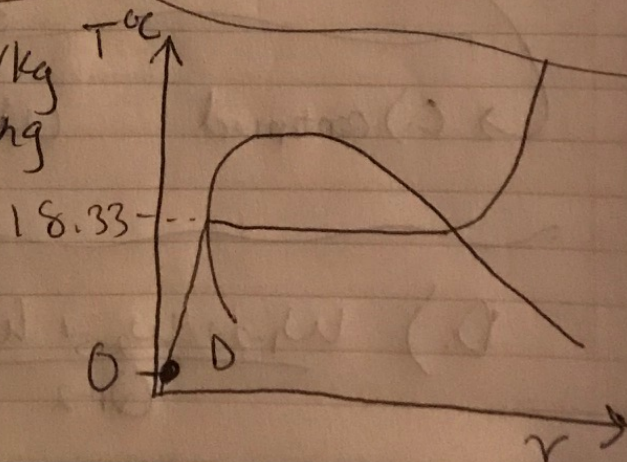
c.) $v = 0.0016386 \text{ m}^3/\text{kg}$
 $h = 274.26 \text{ kJ/kg}$

C = Ammonia at
 20°C



d.) $V = 1.89 \times 10^{-3} \text{ m}^3/\text{kg}$
 $h = 95.1 \text{ kJ/kg}$

D = Propane at 0°C

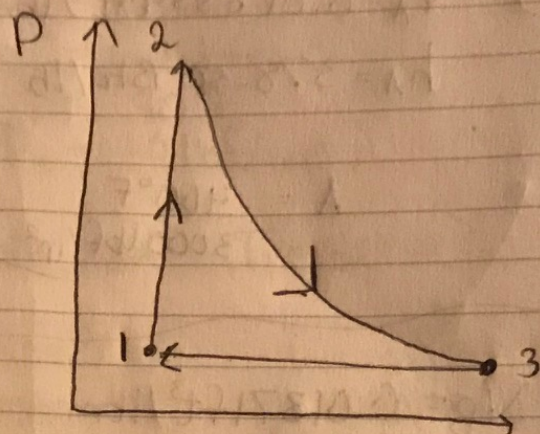


4 4 15 / 15

✓ - 0 pts Correct

- 1 pts $v_a \approx v_f (400^\circ\text{F}) = 0.01864 \text{ ft}^3/\text{lb}$ OR $v_a = 0.018334 \text{ ft}^3/\text{lb}$ (from Table A-5E)
- 1 pts $h_a = h_f (400^\circ\text{F}) + v_f (400^\circ\text{F}) [P - P_{\text{sat}} (400^\circ\text{F})] = 384.6 \text{ Btu/lb}$ OR $h_a = 378.50 \text{ Btu/lb}$ (from Table A-5E)
- 1 pts Using Table A-10E and Eq 3.11, $v_b \approx v_f (95^\circ\text{F}) = 0.01371 \text{ ft}^3/\text{lb}$
- 1 pts Using Table A-10E and Eq 3.14, $h_b \approx h_f (95^\circ\text{F}) = 42.47 \text{ Btu/lb}$
- 1 pts Using Table A-13 and Eq 3.11, $v_c \approx v_f (20^\circ\text{C}) = 1.6386 \times 10^{-3} \text{ m}^3/\text{kg}$
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- 1 pts Using Table A-17 and Eq 3.11, $v_d \approx v_f (0^\circ\text{C}) = 1.89 \times 10^{-3} \text{ m}^3/\text{kg}$
- 1 pts Using Table A-17 and Eq 3.14, $h_d \approx h_f (0^\circ\text{C}) = 95.1 \text{ kJ/kg}$
- 15 pts Missing answer
- 1 pts Missing v.a)
- 4 pts Missing answer for d)
- 2 pts Missing location of points b and c
- 15 pts Missing answer
- 3 pts Wrong answer for a)
- 4 pts Missing answer for c)
- 7 pts Wrong or missing T-v diagram sketches with the location of the points

5.)
3.97



$$T_1 = 800R$$

$$T_2 = 820R$$

$$T_3 = ?$$

a.) $P_1 V = RT_1 \Rightarrow V = \frac{RT_2}{P_2} = \frac{RT_1}{P_1}$

$$P_2 = P_1 \frac{T_2}{T_1} = \frac{20 \cdot 820}{800} \quad \boxed{P_2 = 32.8 \text{ lbf/in}^2}$$

b.) $P_3 = \frac{RT_3}{V_3} = \frac{RT_1}{V_1} \Rightarrow T_3 = T_1 \left(\frac{V_3}{V_1} \right)$

$$T_3 = 800 \cdot 1.432 \quad \boxed{T_3 = 716R}$$

c.) $Q_{12} = W + \frac{m}{M} (u_2 - u_1)$

$$0 + \frac{1}{32} (4119.7 - 2473.2) \Rightarrow \boxed{Q_{1-2} = 51.45 \text{ Btu}}$$

→ c.) continued

$$W_{2-3} + \frac{m}{M} (u_3 - u_2) = \frac{-11}{32} (\Delta u)$$

$$\boxed{W_{2-3} = 17.11 \text{ Btu}}$$

d.) $\frac{W_{1-2} + W_{2-3} + W_{3-1}}{Q_{1-2}} = \frac{17.11 + -13.4}{51.45} = \boxed{7.2\% = \eta}$

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- 0 pts Correct
- 2 pts Wrong or missing p-v diagram with sketch of the cycle
- 5 pts Missing or wrong calculation of Q_{in}
- ✓ - 3 pts Missing or wrong calculation of thermal efficiency of the cycle. Correct answer is 12%
- ✓ - 5 pts Missing or wrong calculation of W_{cycle} or Q_{out}
- ✓ - 2.5 pts Missing calculation of V_3 and V_1 OR V_3 and T_3
 - 1 pts W_{3-1} is negative
- ✓ - 1.5 pts Correct equation but wrong solution for thermal efficiency. Correct answer is 12%
 - 20 pts Missing answer
 - 1 pts $T_3 = 887K$
- ✓ - 2.5 pts Missing or wrong calculation of T_2 , T_3 and P_2
- ✓ - 1.5 pts Did not show conversion to english units
 - 10 pts Correct solution process, but wrong values. The correct answer for part a is 12%.

6.)

3.61

$$C_u C_s = 0.385 \text{ J/g}^\circ\text{C}$$

$$.385 \cdot 13000(T-27) = 4.186 \cdot 4000(50-T)$$

$$T = 44.7^\circ\text{C}$$

$$w = mc = 5005 \text{ J/}^\circ\text{C}$$

$$m = 1195.65 \text{ grams}$$

$$\text{Mass Total} = 1195.65 + 4000 = 5195.65 \text{ g}$$

$$100,000 \text{ J} = 5195.65 \text{ g} \cdot 4.186(T-44.7)$$

$$T = 49.3^\circ\text{C} = \text{Final Temperature of System.}$$

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✓ - 0 pts Correct

- 15 pts Prob 3.61

- 15 pts Final T?

- 5 pts Tf Calculation was incorrect

- 5 pts Specific heat of copper incorrect

- 10 pts Energy balance equation was incorrect